

ROCK,
PAPER,
SCISSORS

LESSON PLANS

The core activities in this unit total about 25.5 hours of lesson time. The Content Descriptions (Australian Curriculum: Science) for each activity are listed here as curriculum codes and described fully in each lesson plan.

Part 1 Why do we use the materials we do?

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
1.1	It's a material world	AC SIS140	Engage	Students are asked to identify the properties of the materials used in the Rock, Paper, Scissors game and explore the properties and used of materials around them.	Core
			Hands-on, Classroom & Digital		
			Medium		
1.2	What is the property?	AC SSU151	Explore & Explain	Students examine groups of materials to determine their common properties and link them to the correct scientific words.	Core
			Hands-on		
			Medium		
1.3	Metals and their properties	AC SSU151	Explore	Students will examine a range of metals to explore their key properties. They will go on to consider ways in which metals are used in modern societies.	Core
		AC SHE 134	Hands-on & Digital		
		AC SHE136 AC SIS141 AC SIS144 AC SIS145 AC SIS234	Medium		
1.4	Rock to scissors	AC SSU151	Elaborate & Evaluate	Students will use the Student Digital to discover what minerals are mined near them and research the properties and uses of iron.	Optional
			Classroom & Digital		
			Medium		

Part 2 What is the matter?

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
2.1	<u>Investigating the properties of solids, liquids and gases</u>	ACSSU151 AC SIS139	Explore	Students will conduct a series of experiments to discover some of the properties of solids, liquids and gases.	Core
			Hands-on		
			Medium		
2.2	<u>Tiny particles</u>	ACSSU151 ACSHE 134 ACSHE226 AC SIS144	Explore & Explain	Students will use a learning object to explore the behaviour of particles in each state of matter and then demonstrate their understanding by using the particle motion to explain the Brownian motion of particles in milk observed under the microscope.	Core
			Hands-on & Digital		
			Medium		
2.3	<u>Slime – what is its state?</u>	ACSSU151	Elaborate	Students make slime (corn flour) and investigate its properties. They will try to classify slime based on what they know about the three states of matter. They will explore this special type of fluid.	Optional
			Hands-on		
			Medium		
2.4	<u>What state is that?</u>	ACSSU151	Explore & Elaborate	Students apply what they have learnt about the three states of matter to everyday materials.	Optional
			Classroom & Digital		
			Medium		
2.5	<u>Gas detective</u>	ACSSU151	Explore & Elaborate	Students conduct a series of inquiries about gases, using household items and then choose one of three topics for further investigation.	Optional
			Hands-on		
			Medium		
2.6	<u>Two more states that matter</u>	ACSSU151	Elaborate	Students discover two more states of matter: plasma and Bose-Einstein condensate.	Optional
			Digital		
			Short		

Part 3 Elements, compounds and mixtures

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
3.1	Is it pure or a mixture?	ACSSU152 ACSIS140 ACSIS141	Engage, Explore, Explain & Elaborate	Students explore the differences between a pure substance and a mixture through some sandcastle play. Students revise their understanding of mixtures, pure substances, elements and compounds and sort substances accordingly.	Core
			Hands-on & Digital		
			Long		
3.2	Separating mixtures	ACSSU152 ACSIS140	Elaborate	Students will use their knowledge of the properties of substances to separate a mystery mixture.	Core
			Hands-on & Digital		
			Short		
3.3	What are alloys?	ACSSU152	Elaborate	Students investigate different useful alloys and determine how their elements affect their properties.	Core
			Digital		
			Short		
3.4	Do the powders contain the same particles?	ACSSU152 ACSIS139	Explore & Evaluate	Students investigate the properties of three copper compounds to discover the differences between elements and compounds.	Core
			Hands-on		
			Long		
3.5	Splitting water	ACSSU152 ACSIS140	Explore	Students electrolyse water using basic equipment.	Core
			Hands-on		
			Medium		
3.6	Elements: the simplest substances	ACSSU152	Explain	Students will explore the range of elements and their properties using an interactive periodic table.	Core
			Digital		
			Short		

Part 3 Elements, compounds and mixtures (continued)

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
3.7	The purity of water	ACSSU152	Explore & Explain	Students use models to learn about the difference between elements, compounds and mixtures.	Core
			Classroom		
			Medium		
3.8	The periodic table	ACSSU152	Elaborate	Students carry out an activity on-line to simulate Mendeleev's analysis of six elements to produce the very first fragment of the periodic table.	Core
			Digital		
			Short		
3.9	Iron – element, compound and mixture	ACSSU152	Elaborate	Students will investigate some uses of the element iron, and compounds and mixtures of iron.	Core
			Classroom		
			Short		
3.10	Is it an element, compound or mixture?	ACSSU152	Evaluate	Students will evaluate their understanding of PART 3 . They will identify elements, compounds and mixtures, and the type and number of elements in a compound.	Core

Part 4 Changing state: a physical change

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
4.1	Observing changes in water	ACSSU151 ACSIS141 ACSIS144 ACSIS148	Explore & Explain	Students investigate the changes that take place when water moves from a solid □ liquid □ gas, and identify the correct terms of those changes. They also identify these changes as examples of a 'physical change'.	Core
			Hands-on & Digital		
			Long		
4.2	Ice play	ACSSU151 ACSHE 135 ACSHE136	Explore	Students follow a recipe for making ice cream. The recipe utilises the effect of salt on the melting point of water to lower the freezing point.	Core
			Hands-on		
			Medium		
4.3	Can matter skip states?	ACSSU151	Engage & Explore	Students see examples of substances that change from a solid, directly to a gas (sublimation). They also learn about uses of sublimation.	Core
			Digital		
			Short		
4.4	Detecting an invisible gas	ACSSU151 ACSSU152	Explore & Explain	Students use a simple test to detect the presence of carbon dioxide gas. They go on to complete a digital activity familiarising them with a number of gases that they may come across in everyday life.	Optional
			Hands-on & Digital		
			Short		
4.5	What is happening to me?	ACSSU151	Elaborate & Evaluate	Students imagine they are water molecules in an ice block and describe what happens to them when left in the sun.	Core
			Classroom		
			Short		

Part 5 Chemical change

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
5.1	Some chemical changes	ACSSU225 ACSIS145	Engage	Students conduct a series of experiments, observing different examples of chemical reactions.	Core
			Hands-on		
			Medium		
5.2	Observing reactions	ACSSU225 ACSIS145	Explore & Explain	Students observe a further range of chemical reactions to extend their knowledge of observations linked to chemical changes.	Core
			Hands-on		
			Medium		
5.3	Reactants and products	ACSSU225 ACSIS140	Explore & Explain	Students conduct a series of experiments to observe chemical reactions. They will identify the reactants and products, and write the word equations for the reactions.	Core
			Hands-on		
			Medium		
5.4	Some natural reactions	ACSSU225	Explain	Students discover some natural reactions – photosynthesis and corrosion.	Core
			Digital		
			Medium		
5.5	Investigating some useful reactions	ACSSU225 ACSHE134 ACSHE136 ACSIS146	Elaborate	Students can make one or more of the following polymers – plastic, glue or rubber.	Core
			Hands-on		
			Long		
5.6	A rusty problem	ACSSU225 ACSIS141	Explore & Explain	Students conduct investigations to answer the question 'What is rust?'	Core
			Hands-on		
			Long		
5.7	Preventing rust	ACSSU225 ACSHE135 ACSIS141	Elaborate & Evaluate	Students conduct an investigation to answer 'How do we prevent the corrosion of iron?' They will also investigate how sacrificial anodes prevent rusting.	Core
			Hands-on		
			Medium		

Part 5 Chemical change (continued)

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
5.8	<u>Do other metals corrode like iron?</u>	ACSSU225	Elaborate	Students design and conduct investigations to answer the question ' <i>Do other metals corrode like iron?</i> '	Core
		ACSIS140	Hands-on		
		ACSIS141 ACSIS145	Long		
5.9	<u>Chemical hazards</u>	ACSHE134	Elaborate	Students watch short video clips to reflect on the impact of toxic particles in the air and consider safety precautions when dealing with chemicals.	Core
		ACSHE135	Digital		
		ACSIS140	Short		
5.10	<u>Sample test</u>		Evaluate	A sample summative test (editable version) and marking scheme are available to teachers by email request from <i>Science by Doing</i> (sbd@science.org.au). In your email request please include your school (or university) and contact details. Note - <i>Science by Doing</i> provides sample assessment items and whilst every effort has been made, the security of these items cannot be guaranteed. <i>Science by Doing</i> encourages teachers to modify the items to suit individual teaching programs.	Optional

1.1 It's a material world

Curriculum links

AC SIS140 Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed.

Lesson outcomes

At the end of this activity students will be able to:

- identify some properties of materials
- describe how the properties of materials are related to their use
- describe the importance of working safely in the science laboratory.

What ideas might your students already have?

- Students have difficulty distinguishing between an object and the material or substances from which it is made.
- Students may not have considered everyday materials in terms of the properties that make them useful. They may focus mainly in easily observable properties such as colour and shape.

Key vocabulary:

Property

Equipment list

Each GROUP will require:

- access to *Student Digital*

Each STUDENT will require:

- *Notebook*

Things to consider and hints for success

- Students feel more confident presenting group ideas, as in *Think-Pair-Share (Step 2)* or via a show of hands (*Step 3*), than when they asked for their individual answers.
- Using thumbs up/thumbs down is even less threatening, as this can be done in front of the body, unseen by the class.
- Students may have already studied laboratory safety so this is an opportunity to revise before commencing laboratory work. Teachers can select appropriate *Notebook* questions to revise and reinforce safety in the laboratory.

Teacher content information:

In Rock, Paper, Scissors, students use their hands to represent each item (see images/photo in *Student Guide*). Each participant shakes a fist three times and on the third shake forms one of the three shapes with their hand. If both participants form the same shape it is a draw. If students form different shapes, the following rules apply:

1. rock beats scissors, because rock blunts scissors
2. scissors beat paper, because scissors cut paper
3. paper beats rock, because paper wraps up rock. The winner is the first person to win three times.

Rock, Paper, Scissors

1



▲ As a teacher, it is your responsibility to follow your school safety and risk, policies and procedures.

▲ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

▲ Always ensure safe and ethical disposal of chemicals.

▲ Use teacher judgement when considering equipment substitutions.

▲ Make considered choices selecting and using resources to minimise waste.

Lesson plan

Step 1: Ask students who have played Rock, Paper, Scissors to raise their hands.

Step 2: Students use a **Think-Pair-Share** strategy to discuss and complete **Steps 1** and **2**.
Use a show of hands to share pairs' ideas with the class.

Answer: (a) **rock** crushes scissors, (b) **paper** wraps around rock, (c) **scissors** cut paper

It is possible to come up with some alternative pairings based on other characteristics. For example, rock beats paper because it is stronger, but explain to students that the game is based on one particular set of agreed characteristics.

Step 3: Students play with their partner. Remind them the winner is the first person to win three times.

Step 4: Students answer the discussion question in their **Notebooks**, individually, in pairs or as a class.

Step 5: Students work individually to complete the second part of this activity. They consider how the properties of materials relate to their uses.

Step 6: Students share answers with their partner and/or the class.

Step 7: Students move to digital component to revise lab safety and complete **Notebook** questions.



△ As a teacher, it is your responsibility to follow your school safety and risk, policies and procedures.

△ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

△ Always ensure safe and ethical disposal of chemicals.

△ Use teacher judgement when considering equipment substitutions.

△ Make considered choices selecting and using resources to minimise waste.

1.2 What is the property?

Curriculum links

ACSSU151 Properties of the different states of matter can be explained in terms of the motion and arrangement of particles.

KEY IDEAS

- Different substances have different properties, allowing them to be used for different purposes.

Lesson outcomes

At the end of this activity students will be able to:

- use appropriate scientific terms to describe the properties of materials.

What ideas might your students already have?

Students have difficulty identifying that everything around them is a chemical substance with particular properties.

Key vocabulary:

Viscosity, elasticity, hardness, plasticity, malleability, ductility.

Equipment list

Each CLASS will require

- Six collections of substances (five or six items in each) demonstrating each of the following properties:
- viscosity (e.g. shampoo, honey, water, milk, pouring cream)
- elasticity (e.g. rubber band, spring, rubber tyres from a toy car, plastic ruler, rubber bike tube)
- hardness (e.g. glass bottle, scissors, drill bit, fork, a rock, a pair of pliers, metal hammer)
- plasticity (e.g. cane basket, play dough, clay, plasticine, bread dough, slime)
- malleability (e.g. lead sinker, copper pipe/saucepan, metal jewellery, aluminium foil or can)
- ductility (e.g. metal coat hangers, various wires, garden basket hooks, paper clips and electrical wires).
- Lustre (e.g. polished metal jewellery, cutlery, aluminium foil, coins, medallions)

Each STUDENT will require:

- **Notebook**

Things to consider and hints for success

This activity is a caucus race. There are six stations, so the class is divided into six groups or into 12 groups which use two separate set-ups.

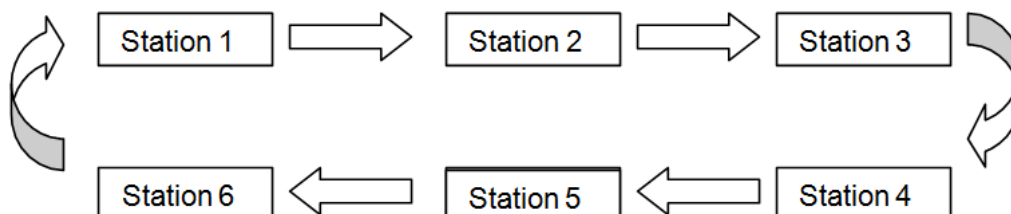
Caucus race

Each group rotates around the stations (collections) under the teacher's direction. It is important to first explain the rules to students.



1. Allocate groups to their starting stations.
2. Tell them they will have three minutes to record their observations.
3. Then direct groups move to the next station in the sequence.

e.g.



Lesson plan

- Step 1:** Form groups and explain how the race works. Students move around the stations, listing the common properties of all the materials in their **Notebooks**.
- Step 2:** The groups answer the discussion question in their **Notebooks**.
- Step 3:** Students share the properties for each set with the class. Reach a class consensus for each set and record the results on the board in a table (see below). Correct scientific terminology is not yet expected.

Table:

Set	Properties	Examples

Using a table means this information can be used to complete the next part. Only one column will need to be changed.

- Step 4:** Students complete the second part of the activity. This can be done individually, in groups or as a class, depending on the ability of the group or time available. Use the table on the board (from **Step 3**) to either work as a class or to discuss group responses.
- Step 5:** The discussion could be done individually as a **formative assessment** of students' ability to use the correct terminology to identify the properties of materials.



1.3 Metals and their properties

Curriculum links

- ACSSU151 Properties of the different states of matter can be explained in terms of the motion and arrangement of particles.
KEY IDEAS
- Different substances have different properties, allowing them to be used for different purposes.
 - Metals have a range of special properties that make them particularly useful in many applications.
- ACSHE 134 Scientific knowledge has changed peoples' understanding of the world and is refined as new evidence becomes available.
- ACSHE136 People use science understanding and skills in their occupations and these have influenced the development of practices in areas of human activity.
- AC SIS141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.
- AC SIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.
- AC SIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.
- AC SIS234 Use scientific knowledge and findings from investigations to evaluate claims based on evidence.

Lesson outcomes

At the end of this activity students will be able to:

- describe a range of properties common to metals
- recognise the importance of metals to their daily lives.

Key vocabulary:

Conductivity, lustre

Equipment list

Each GROUP will require:

- several samples of different metals and non-metals
- a simple electrical circuit with globe to test conductivity
- hand lens or dissecting microscope
- emery paper
- hammer

Each STUDENT will require:

- **Notebook**



Things to consider and hints for success

Students may need to be reminded about the basic operation of an electrical circuit, depending on their previous experiences.

The digital activity *Down to Earth: metals matter* (L927) in the **Notebook** is housed in *Scootle*. You (as educator) will need to register on *Scootle* (<https://www.scootle.edu.au/ec/p/home>) in order to access this resource for your students. Once registered you can search for the resource using its ID code L927 and add the resource to your own learning path. Click on the 'Add to' button, and choose 'Create new learning path'. Once you have curated the required resources you need in a learning path, click on 'Learning paths' at the top of the page, open your learning path and copy the six letter PIN number that you can share with your students. Your students need to go to the *Scootle* home page (<https://www.scootle.edu.au/ec/p/home>), click on 'Student login' at the top of the page and enter the PIN number.

Teacher content information:

This activity introduces students to the properties of metals before dealing with elements and compounds. Strictly speaking the distinction between metals and non-metals is restricted to the elements. However, many alloys, such as brass and steel, are regarded as metals in common usage since they share all the essential properties with metallic elements. The focus in this activity is on the properties associated with metals and how these make metals one of the most useful types of materials in the modern world. The mining of metallic ores in Australia accounts for more than 100 billion dollars of Australia's annual GDP (and around a quarter of our export earnings). Metals really are important to Australia.

An additional property of metals that is not specifically explored in this activity is conductivity of heat. In fact the conductivity of electricity and heat are both mediated by the de-localised electrons so that these two properties tend to match.

Lesson plan

Step 1: Students observe each sample. If the sample is not shiny they try polishing it with the emery paper.

Step 2: Students test each sample using some of the tests you learnt about in **Activity 1.2**. This includes a test of whether the sample is able to conduct electricity. They test for malleability using a hammer. Explain how the test for electrical conductivity works, depending on the past experience of the class.

Step 3: Observations are recorded in a table in their **Notebook**.

Discussion:

At the end spend some time discussing what life would have been like before the discovery of how to produce pure metals.

1. Which group (metals or non-metals) had properties that were most similar?
2. Describe some of the properties that you think make metals very useful to us.



1.4 Rock to scissors

Curriculum links

ACSSU151 Properties of the different states of matter can be explained in terms of the motion and arrangement of particles.

KEY IDEAS

- Metals have a range of special properties that make them particularly useful in many applications.

Lesson outcomes

At the end of this activity students will be able to:

- name some minerals mined near them
- describe some properties and uses of iron.
- identify the manufacturing process used to produce a pair of scissors from a rock.

What ideas might your students already have?

Depending on the region where your students live, some students will have advanced knowledge of mining activities while others will have limited knowledge

Key vocabulary:

Ore, metal, shiny, mixture, alloy, liquid, solid, melting point, mining, hard, malleable, manufacture, mould.

Equipment list

Each GROUP will require:

- access to **Student Digital**

Each STUDENT will require:

- Notebook**

Things to consider and hints for success

- You may need to demonstrate how to draw a flow chart to summarise ideas. (If required, students may be given some terms to use in their flow chart).
- Students use *Find out more* to research the properties and uses of iron. This can then be followed by a more general discussion on mining activities in their local area or state to promote interest and understanding of the benefits and issues associated with mining. Has anyone visited a mining site?
- When using the *Australian Mines Atlas* to find out what is mined nearby, first check the steps below, as the site may have been updated.
- Each time you make a selection, use 'Refresh Map'.

Finding your location

- select your state or territory on the bottom tool bar
- select Map Layers tab
- select Geographic layers folder
- select Population centers
- use zoom tool to find here you live.



Finding mines near you

- select Mineral Deposits and Occurrences folder
- select Operating Mines – All commodities
- Find the name of the mine(s) nearest you.

Finding the mineral(s) mined near you

- select Resources Search tab
- type in name
- search.

To view the interactive history in *Find out more*, Google Earth must be installed.

Teacher content information:

Iron (Fe) is the fourth-most abundant element in the Earth's crust and exists in nature as iron ore. Australia is the world's second-largest producer of iron ore after China, with 98 per cent from the Pilbara region of Western Australia. The remainder of Australia's production comes from mines in Tasmania, New South Wales, Queensland and South Australia.

Rock containing ore (solid) → mined → iron extracted from ore → iron metal (solid, malleable and shiny) → heated until it reaches melting point (liquid) → mixed with other metals and non-metals to form alloy (mixture) → poured into mould to manufacture scissors → cooled → scissors (solid and hard).

Lesson plan

Step 1: Ask students to look at the images of the rock and scissors and roll over images to see how the rock became scissors. This leads students to explore minerals mined in Australia.

Step 2: Students use the *Australian Mines Atlas* to find minerals mined nearby. Instructions on using this resource are found in **Things To Consider And Hints For Success**.

Step 3: Students use the links to digital resources to find five interesting facts about iron. (These facts will be useful in the *Information report* assessment task).

Step 4: Students answer six questions.

Step 5: Students prepare a flow chart to summarise the manufacturing process: rock → iron → scissors.

Find out more:

Students explore the history and importance of mining in Australia, and can view more information about iron ore mining. This information could be used as part of their research into the properties and uses of iron.



2.1 Investigating the properties of solids, liquids and gases

Curriculum links

ACSSU151 Properties of the different states of matter can be explained in terms of the motion and arrangement of particles.

KEY IDEAS

- All substances are made of particles, which are in constant motion. These particles are usually atoms and molecules.
- The particle model can be used to explain a range of phenomena, including changes of state, and physical and chemical changes.

ACSIS139 Identify questions and problems that can be investigated scientifically and make predictions based on scientific knowledge.

Lesson outcomes

At the end of this activity students will be able to:

- identify some properties of solids, liquids and gases.

What ideas might your students already have?

Students might be able to give examples of substances in each state. They should also be able to suggest some properties of each state, in their own words, from the previous activity. They may still have difficulty recognizing air and other gases as materials or substances.

Equipment list

Each GROUP will require:

- wooden block (if possible needs to fit within syringe and beakers)
- 100 mL water in an measuring cylinder
- 100 mL beaker
- 250 mL beaker
- large syringe

Each STUDENT will require:

- **Notebook**
- Safety glasses

Things to consider and hints for success

Students should first complete all tests on the solid, and then repeat the same tests for the liquid and gas. You may have to guide them with the air, as they cannot see volume or shape. You could get them to predict and then discuss what they would observe if they could see the air. Then get them to explain it. This could be done as a class after they have completed solids and liquids.

When doing **Test 3** on the liquid, it is important to point out they must remove all air from the syringe, otherwise it may look like it is compressing because the air in the cylinder is compressing. To remove the air, fill the syringe with water by drawing it into the syringe, then tip the syringe upside down and push the plunger until water starts to dribble out.



So you don't give away the answer to the questions, explain to students: *Make sure you remove all the air from the syringe when testing the liquid. If you don't, you will be testing a gas and a liquid and this will affect your observations.*

Students may complete a *Risk Assessment* before commencing this activity.

Teacher content information:

This activity uses the **Predict-Observe-Explain** strategy:

Predict – before students start each test they should predict what might happen and record it in their **Notebooks**.

Observe – students conduct the test and record their observations.

Explain – students write an explanation for their observations.

Lesson plan

Step 1: Students work through each test on solids liquids and gases, recording their results in their **Notebooks**.

Step 2: Discuss their results as a class.

Follow up:

You could ask students to do a summary table of their/the class results.



2.2 Tiny particles

Curriculum links

- | | |
|-----------|---|
| ACSSU151 | Properties of the different states of matter can be explained in terms of the motion and arrangement of particles
KEY IDEAS <ul style="list-style-type: none"> • All substances are made of particles, which are in constant motion. These particles are usually atoms and molecules. • The particle model can be used to explain a range of phenomena, including changes of state, and physical and chemical changes. |
| ACSHE 134 | Scientific knowledge has changed peoples' understanding of the world and is refined as new evidence becomes available. |
| ACSHE226 | Science knowledge can develop through collaboration across the disciplines of science and the contributions of people from a range of cultures. |
| AC SIS144 | Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate. |

Lesson outcomes

At the end of this activity students will be able to:

- use the kinetic theory (particle model) to explain the behaviour of particles in each state of matter.
- use the particle model to explain the phenomenon of Brownian motion.

What ideas might your students already have?

Students will be familiar with the concept of the particle model, but have had very few opportunities to see direct evidence of it.

Equipment list

Each **GROUP** will require:

- access to **Student Digital**
- microscope, slides and cover slips
- small quantity of milk diluted approximately 1 in 4.
- pipette
- tissues for drying slides

Each **STUDENT** will require:

- **Notebook**

Things to consider and hints for success

Milk is an emulsion of mostly droplets suspended in water. The fat droplets have a variety of sizes that are appropriate for the observation of Brownian motion. Observing Brownian motion is difficult because it requires the compound microscope to be well focused under 400 X magnification. Students consistently have problems getting microscopes to work to this level, and may need some direct assistance in setting this up.



The phenomenon is quite underwhelming to students until you impress on them that the small jiggling of particles that they observe is due to collisions with water molecules in the milk. As they get used to the phenomenon it becomes more obvious. It should be possible to focus on a couple of specks that are close to each other and observe them wandering around in a 'random walk' due to their random collisions with the water molecules around them.

There is something of a misconception that Einstein received one of two Nobel Prizes for his analysis of Brownian motion. However, he only received one Nobel prize (for his analysis of the photoelectric effect).

Teacher content information:

The particle model (or kinetic theory) of matter is one of the most important scientific ideas that students learn. And yet it is one of the most difficult to demonstrate directly at a macroscopic level. This activity provides an opportunity for students to consider the particle model and its ramifications, as well as making some direct observations that provide evidence of particles in constant motion.

The position and motion of particles in different states.

State	Position and motion of particles
Solid	The particles are in fixed positions in a lattice structure. They are tightly packed together and can only vibrate within a particular position.
Liquids	Particles clump together but are not locked into fixed positions. The clumps are always large enough to fall to the bottom of any container due to gravity. They are close together.
Gases	Particles move quickly. They travel in straight lines until they hit another particle. They are widely spaced and do not interact except through collisions.

Brownian motion

Einstein's mathematical analysis of Brownian motion was important in providing direct evidence for the particle (kinetic) theory of matter. Although not as well-known as his work on relativity and the photoelectric effect this research was one of the important pieces of early work that propelled him into a career as a theoretical physicist.

Lesson plan

Step 1: Students observe the movement of particles in each state of matter using the learning object, then answer Question 1 in their **Notebooks**.

Step 2: Students complete the remaining questions in their **Notebooks**.

Step 3: Students then move on to the activity of observing Brownian motion in milk. Spend time ensuring that students are able to focus on their slides using 400X magnification.



Discussion Questions

You should observe the fatty droplets in the milk jiggling around.

1. You will notice both large and small droplets. Do you observe any difference between the jiggling motion of the large and small droplets?
2. Why do you think there is a difference?
3. How do your observations support the idea that water is made of tiny particles in constant motion?



⚠ As a teacher, it is your responsibility to follow your school safety and risk, policies and procedures.

⚠ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

⚠ Always ensure safe and ethical disposal of chemicals.

⚠ Use teacher judgement when considering equipment substitutions.

⚠ Make considered choices selecting and using resources to minimise waste.

2.3 Slime – what is its state?

Curriculum links

ACSSU151 Properties of the different states of matter can be explained in terms of the motion and arrangement of particles.

KEY IDEAS

- All substances are made of particles, which are in constant motion. These particles are usually atoms and molecules.
- The particle model can be used to explain a range of phenomena, including changes of state, and physical and chemical changes.

Lesson outcomes

At the end of this activity students will be able to:

- identify the state of a substance that doesn't behave as expected
- give everyday examples of where fluids don't behave as expected.

What ideas might your students already have?

Students will be familiar with the general properties of the three states of matter and be able to identify the states of everyday materials.

Equipment list

Each GROUP will require:

- 1 cup (250 mL) water
- 1 cup of cornflour*
- a container (e.g. ice-cream container)
- stirring rod or spoon
- food colouring (optional)**

Each STUDENT will require:

- **Notebook**
- Access to **Student Digital**

Things to consider and hints for success

*This activity will only work using cornflour from corn/maize. Read the pack carefully, as most supermarket 'cornflour' is made from wheat.

**Take care, as too much food colouring will stain hands.

Students should complete a *Risk Assessment* before commencing this activity.

Teacher content information:

Fluids which flow easily are said to have low viscosity (e.g. water), whereas fluids which do not are said to have a high viscosity (e.g. cold honey). Cornflour slime is a special type of fluid that doesn't follow the usual rules of fluid behaviour. When pressure is applied to slime, its viscosity increases and the cornflour slime becomes thicker. At a certain point, slime actually seems to lose its flow and behave like a solid. Cornflour slime is an example of a shear-thickening fluid.



The opposite happens in shear-thinning fluids; they get runnier when you stir, shake or compress them. For example, when toothpaste is sitting on a toothbrush it is so thick you can turn the brush upside down and the toothpaste doesn't drip off. But if it was that thick when you tried to squeeze it out of the tube, you would not be able to. Fortunately, toothpaste gets runnier when you squeeze (compress) it. Other shear-thinning fluids include blood, paint, ballpoint pen ink, nail polish.

Lesson plan

Step 1: Groups make slime then answer the discussion question.

Discussion question:

What substance test could you use to help you classify slime?

Step 2: Students work through the activity to find an explanation for their observations.

Step 3: Students complete the activity by researching Newtonian and Non-Newtonian fluids.



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△ Always ensure safe and ethical disposal of chemicals.
△ Use teacher judgement when considering equipment substitutions.
△ Make considered choices selecting and using resources to minimise waste.

2.4 What state is that?

Curriculum links

ACSSU151 Properties of the different states of matter can be explained in terms of the motion and arrangement of particles.

KEY IDEAS

- The particle model can be used to explain a range of phenomena, including changes of state, and physical and chemical changes.

Lesson outcomes

At the end of this activity students will be able to:

- identify the states of everyday materials.

What ideas might your students already have?

Students should know the properties of the three states of matter and be able to identify the states of everyday materials.

Equipment list

Each CLASS will require:

- access to **Student Digital**
- jar of granulated sugar
- ingredients for elephant toothpaste

Each STUDENT will require:

- Notebook**

Things to consider and hints for success

You might like to start the activity by demonstrating elephant toothpaste. There are several recipes on the internet. These are either for teacher demonstrations or student-friendly experiments. Some may contain chemicals now banned in schools. Check with your laboratory technician, advisor or your education authority if you are unsure.

Video clips at the end of this activity demonstrate elephant toothpaste.

Teacher content information:

Elephant toothpaste is made using the decomposition of hydrogen peroxide. Dishwashing liquid is added to make the solution foam when oxygen gas is released. A catalyst is added to rapidly breakdown the peroxide.

Students are only required to write the word equation in this activity. You could get them to write the unbalanced chemical equation to show how symbols are used – depending on the ability of the class.

Reactant = hydrogen peroxide (H_2O_2 (l))

Products = oxygen gas (O_2 (g)) and water (H_2O (l))

(NOTE: technically you should use (aq) for water, however at this stage we are only asking students to identify substances as solids, liquids or gases).



Word equation: hydrogen peroxide (l) → oxygen (g) + water (l)

Balanced equation: $2\text{H}_2\text{O}_2 (\text{l}) \rightarrow \text{O}_2(\text{g}) + 2\text{H}_2\text{O} (\text{l})$

Lesson plan

Step 1: Demonstrate elephant toothpaste (optional).

Step 2: Students look at each substance on the sick elephant's hospital tray and identify its state. This could be done as a class activity using an interactive whiteboard or data projector. Use a show of hands or the less threatening thumbs up/thumbs down:

- thumb up (solid)
- thumb horizontal (liquid)
- thumb down (gas).

This method is less threatening as it can be done in front of the body, unseen by the class.

Step 3: Show students a jar of granulated sugar. Get them to answer Questions 1 and 2 in their **Notebooks**. Discuss their answers.

Step 4: Students watch videos on elephant's toothpaste then answer Questions 3 and 4. You might need to do this as a class or optional extension activity depending on student ability and familiarity with chemical equations. At this stage, they are only being introduced to the terms chemical equation, reactants and products.



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2.5 Gas detective

Curriculum links

ACSSU151 Properties of the different states of matter can be explained in terms of the motion and arrangement of particles.

KEY IDEAS

- The particle model can be used to explain a range of phenomena, including changes of state, and physical and chemical changes.

Lesson outcomes

At the end of this activity students will be able to:

- apply the properties of gases to different situations.

Equipment list

Each GROUP will require:

- Activity Sheet 2.8 Gas Detective Instructions.

Inquiry 1:	Inquiry 2:	Inquiry 3:	Inquiry 4:	Inquiry 5:
- balloon - conical flask* - teaspoon - bicarbonate of soda 50 mL measuring cylinder - vinegar.	bottle of perfume or strong smelling liquid in a jar (e.g. ammonia) label the jar <i>perfume</i>.	- balloon - candle - watch glass or base for candle - box of matches.	- hard-boiled eggs, shelled - conical flask** - pieces of folded paper for burning.***	- balloon - conical flask* 2 beakers**** - iced water - warm water.

Each STUDENT will require:

- Notebook**

Things to consider and hints for success

Give each group the equipment for each inquiry in a separate container with its instructions. Alternatively, conduct a caucus race.

* The neck of the conical flask needs to be small enough to allow the balloon to fit over it.

** The neck of the conical flask needs to be wide enough for the egg to just sit on it, without slipping into the flask.

*** A strip of paper, about ½ A4 in size, folded lengthways to a width of about 2 cm, then twisted. Any scrap paper can be used.

**** The beaker needs to be large enough for the conical flask to sit in it.



Familiarise yourself with each test beforehand especially if you are improvising with different equipment or substances.

Students should complete a *Risk Assessment*.

Teacher content information:

Inquiry	Explanation
1	Carbon dioxide gas is produced. It takes up space and is lighter than air and so rises, filling the balloon.
2	The perfume evaporates and travels through the air by diffusion. Diffusion is the movement of particles from an area of high concentration of that particle (the perfume bottle) to areas of low concentration of that particle (the classroom).
3	The water in the balloon is absorbing the heat from the candle. The balloon conducts heat very well, so the flame heats the water without harming the balloon.
4	When you drop burning paper into the bottle, the paper will burn until all the oxygen is consumed. The burning paper heats the air in the bottle, increasing the air pressure. The heated air pushes the egg out of the way, making it appear to jump on the mouth of the bottle. As the air cools, the egg settles down and seals the mouth of the bottle. There is now less air in the bottle than when you started, and therefore a lower air pressure. Since the air pressure inside the bottle is lower than outside, the outside air exerts a force on the egg, pushing it into the bottle.
5	When the air in the flask cools, the particles move closer together, making the balloon shrink. When the air in the flask heats, the particles spread further apart. The balloon blocks the expanding air from escaping, therefore the balloon increases in size.

Lesson plan

Step 1: Groups conduct each inquiry using the **Predict-Observe-Explain** strategy. Ensure they first read the instructions.

If you are using a chemical other than perfume for Inquiry 2, ensure students are informed of any safety requirements.

Step 2: As a class discuss student observations and explanations.

Step 3: In pairs, students continue their investigations of gases by selecting one of three activities to research at home.

- methane gas
- kitchen chemistry
- diffusion of gases.



2.6 Two more states that matter

Curriculum links

ACSSU151 Properties of the different states of matter can be explained in terms of the motion and arrangement of particles.

KEY IDEAS

- The particle model can be used to explain a range of phenomena, including changes of state, and physical and chemical changes.

Lesson outcomes

At the end of this activity students will be able to:

- list five states of matter
- identify some uses of plasma and Bose-Einstein condensate (BEC).

What ideas might your students already have?

Students have learnt about the three states of matter – solids, liquids and gases and may have heard of plasma.

Key Vocabulary:

Plasma, Bose-Einstein condensate, absolute zero, postulate.

Equipment list

Each CLASS will require:

- access to **Student Digital**

Each STUDENT will require:

- Notebook**

Things to consider and hints for success

Explain that students will find the answers to the questions in the videos. They could first write the questions in their **Notebooks**.

Ensure students are familiar with the temperature ranges illustrated on the thermometer in the student guide prior to watching the BEC video.

Teacher content information:

This activity (or part thereof) could be used any time after **Activity 2.1** as an extension or as homework.

Plasmas are particularly important, since the sun (and hence most of the solar system by mass) is made of plasma rather than a gas. When we say that the sun is comprised of hydrogen we mean that it is made mostly of a plasma of individual protons and electrons. Only on a planet, such as Earth, where conditions are relatively cool and dense, can atoms form and survive.

Lesson plan

Step 1: Students investigate extreme ranges in temperature using the reference material and questions in the Student Guide.

Step 2: Students work through the digital activity, including *Find out more*, recording their answers to questions in their **Notebook**.



3.1 Is it pure or a mixture?

Curriculum links

- ACSSU152 Differences between elements, compounds and mixtures can be described at a particle level.
KEY IDEAS
- Mixtures contain different substances mixed together in varying proportions. They can be separated using a range of physical separation techniques.
- ACSIS140 Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed.
- ACSIS141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.

Lesson outcomes

At the end of this activity students will be able to:

- identify some differences between a pure substance and a mixture.

What ideas might your students already have?

Students misunderstand the meaning of the term 'pure'; they might refer to something that is clean and not contaminated.

Key Vocabulary:

Pure substance, mixture, matter, elements, compounds.

Equipment list

Each GROUP will require:

- a tray of sand
- polystyrene or plastic cups
- digital scales

Each STUDENT will require:

- Notebook**
- Internet access

Things to consider and hints for success

- Consider how your students will do strength tests on their sandcastles. You might prefer to use 50 gram masses or large washers instead of coins.

Lesson plan

Step 1: Project page 27 onto the class screen and ask student groups to decide which three ratios of sand to water mixtures they will test.

Step 2: Ask students to consider how they will conduct fair tests when it comes to measuring the strength of each sandcastle.

Step 3: Encourage students to document their investigation and take photos to record each trial.



Step 4: After trialling, bring the class together to discuss the questions on page 27.

Step 5: Invite students to explore the digital activities. The Matter Key is revised and students are asked to sort substances into mixtures and pure substances, then into elements and compounds.

Follow-on:

Students that explore the *Find out more* section might like to follow the suggested advice for building a very strong sandcastle.



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⚠ Make considered choices selecting and using resources to minimise waste.

3.2 Separating mixtures

Curriculum links

- ACSSU152 Differences between elements, compounds and mixtures can be described at a particle level.
KEY IDEAS
- Mixtures contain different substances mixed together in varying proportions. They can be separated using a range of physical separation techniques.
- ACSIS140 Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed.

Lesson outcomes

At the end of this activity students will be able to:

- describe techniques (evaporation, filtration, magnetism) used to separate materials.

What ideas might your students already have?

Separation techniques may have been introduced at Primary School through the *Primary Connections* program.

Equipment list

Each **GROUP** will require:

- a mystery substance (sand, salt, iron filings, mixed)
- Bunsen burner
- paper/tray for tipping mixture on/into
- gauze mat
- protective mat
- evaporating basin
- 250 mL beaker
- filter funnel, stand and paper
- magnet
- matches

Each **STUDENT** will require:

- Notebook**
- safety glasses

Things to consider and hints for success

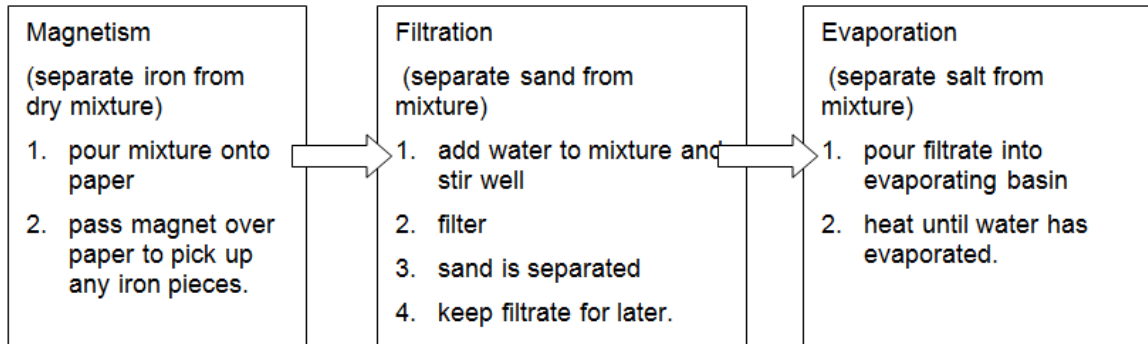
Depending on the class it may be necessary to include some class discussion to remind them of the different separation techniques available.



Teacher content information

Suggested separation techniques (NOTE: students may choose other methods).

Process plans:



Lesson plan

Step 1: Students look at five mixtures (one is a mystery) and watch the videos demonstrating separation techniques.

Step 2: Groups complete the **Notebooking** activity.

Step 3: As a class, discuss students' results.



3.3 What are alloys?

Curriculum links

ACSSU152 Differences between elements, compounds and mixtures can be described at a particle level.

KEY IDEAS

- Mixtures contain different substances mixed together in varying proportions. They can be separated using a range of physical separation techniques.

Lesson outcomes

At the end of this activity students will be able to:

- identify different alloys and explain, using examples, why we use them.
- explain why alloys are mixtures.

What ideas might your students already have?

Students are unlikely to be familiar with the concept of alloys.

Equipment list

Each **GROUP** will require:

- access to **Student Digital**

Each **STUDENT** will require:

- Notebook**

Teacher content information

Some commonly used alloys:

Alloy	Elements	Properties
brass	copper and zinc	harder, more attractive
bronze	copper and tin	harder, easier to melt and cast
cast iron	iron, carbon (2-4%) and silicon	brittle and hard
carbon steel	iron, carbon (0.6-0.8%) (and silicon)	stronger and harder
pewter	tin, antimony and copper	malleable, resistant to corrosion.
solder	lead and tin	melts at a lower temperature
sterling silver	silver and copper	stronger
stainless steel	iron, chromium and sometimes nickel	resists corrosion
wrought iron	iron and carbon (0.2%) and any number of other elements (originally from the slag in the furnace)	harder but still soft enough to shape easily.



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NOTE: the elements contained in the alloys may vary, depending on the source. Those listed above are the alloy's main elements. The percentages of carbon in the steel also vary, depending on the source of the information.

The overall classification of an alloy can be difficult. Since they are not elements they are not classified as metals, but in common usage they would all be described as metals because they share most properties with them. Although alloys are classified as mixtures the different atoms within an alloy are closely bound, often through the same metallic bonding typical of pure metals, and they are not always easy to separate.

Lesson plan

Step 1: Discuss the ideas presented in the Student Guide with the class, with particular reference to the historical importance of alloys, and how bronze had properties different to other metals known at the time.

Step 2: Students look at everyday items made of alloys.

Step 3: They use the learning object to investigate alloys.

Step 4: They use reference materials to answer the two **Notebook** questions.



3.4 Do the powders contain the same particles?

Curriculum links

- ACSSU152 Differences between elements, compounds and mixtures can be described at a particle level.
KEY IDEAS
- A compound is a pure substance that contains more than one kind of atom.
- ACSIS139 Identify questions and problems that can be investigated scientifically and make predictions based on scientific knowledge.

Lesson outcomes

At the end of this activity students will be able to:

- name an element and its compounds.
- describe some chemical tests.

What ideas might your students already have?

If introducing the concept of aqueous solutions note that students sometimes confuse the processes of dissolving and melting.

Key Vocabulary:

Element, compound, control

TEACHER DEMONSTRATION ONLY

Equipment list

Each CLASS will require:

- a jar of each of the substances, A-C, to be tested.

Each CLASS will require:

- substance A (copper sulfate) - substance B (copper chloride) - substance C (copper nitrate) 4 test tubes - test tube rack.	- hand lens or magnifying glass 4 rubber stoppers to fit test tubes 4 iron nails - steel wool.	- paper towel - depression tile - dropper bottle of nitric acid - small piece of copper.
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Each STUDENT will require:

- Science by Doing Notebook*
- safety glasses



Things to consider and hints for success

Planning the demonstration

This activity will need to be completed over a couple of lessons. Test tubes in Part B need to be left overnight (at least). It doesn't matter if they are left longer. Parts A and B are completed in one lesson and then complete Part C in the following lesson. It is recommended you complete a *Risk Assessment* with your students.

Disposal of small quantities of copper solutions

Immerse steel wool in the solution, and thus displace the copper ions. Decant the supernatant from the precipitated copper and flush down the sink. Dispose of the copper metal residue as general waste.

Teacher content information

Copper sulfate, copper chloride and copper nitrate all look similar and their solutions all react with iron to produce copper solid. When the copper compounds are dissolved in water, they separate into particles (ions) (e.g. copper and sulfate particles break apart). When an iron nail is added to the solution, the copper particles attach to the iron, coating the nail in copper. The remaining particles (e.g. sulfate) are left in the solution.

Word equations

Copper sulfate (l) → copper (s) + sulfate particles (l)

Copper chloride (l) → copper (s) + chloride particles (l)

Copper nitrate (l) → copper (s) + nitrate particles (l)

It is technically incorrect to use (l) in these equations. You could introduce aqueous solutions at this point depending on the ability of your students. If you introduce aqueous solutions, replace the (l) with (aq). An aqueous solution is formed when a substance is dissolved in water and breaks into its separate particles (ions). In this unit we are just trying to get students used to using convention when writing formulae, for future chemistry studies.

In Part B, Test tube D is the *control*. A control is the same in every way to the other set-ups, except for the variable being tested. A control is used for a comparison to see if anything has changed. If it has, it can be assumed it occurred due to the variable being tested. In this activity, all the test tubes contain water and Test tubes A-C also contain a copper compound. Therefore if any change takes place in Test tubes A-C, and not D, it can be assumed this change is due to the copper compounds.

Lesson plan (refer also to the Student Guide)

Part A: Do they look the same?

- Show students the jars of three compounds being investigated. Explain the substances are all compounds because they contain more than one kind of atom (element).
- Students make observations about the appearances of the three substances and answer the discussion question.



Part B: How do the substances react with iron nails?

- Teacher or students set up Part B and leave overnight. In the following lesson, they record their observations and answer the discussion questions. They are asked to predict what the brownish-red substance is.

Part C: Testing your prediction.

- Students make observations and answer the discussion questions.
- Students share their findings with the class.

Suggested question:

Do the three substances you started with contain the same kind of particles? If so, what sort?

You could write the formula for each compound on the board to show its elements. You can explain what the numbers mean, as this is what the students will learn at the beginning of the next activities.

- Copper sulfate CuSO_4
- Copper chloride CuCl_2
- Copper nitrate CuNO_3



3.5 Splitting water

Curriculum links

- ACSSU152 Differences between elements, compounds and mixtures can be described at a particle level.
KEY IDEAS
- A compound is a pure substance that contains more than one kind of atom.
- ACSIS140 Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed.

Lesson outcomes

At the end of this activity students will be able to:

- describe how electricity can be used to split water into two gases.

What ideas might your students already have?

Students may consider water to be a fundamental substance that cannot be split into component substances.

Equipment list

Each **GROUP** will require:

- 9 volt battery or power pack
- 2 leads with crocodile clips
- Paper cup
- Scissors
- 2 metal paper clips
- Pin
- Sticky tape
- 2 bulldog clips, 2 rubber bands
- 2 medium test tubes
- 200 ml beaker, stirring rod
- sodium bicarbonate
- matches (optional)

Each **STUDENT** will require:

- *Science by Doing* **Notebook**

Things to consider and hints for success

- The beauty of this activity is that each group of students can conduct the experiment easily at their bench and observe the generation of hydrogen and oxygen at close quarters.
- You may wish to demonstrate this reaction to the whole class using a voltameter (or coulometer) with dilute sulfuric acid rather than sodium bicarbonate.
- Care needs to be exercised when using a 9V battery to avoid a small electric shock.



Teacher content information

An electrolyte (sodium bicarbonate) is added to the water to overcome some high activation potentials involved in the electrolysis of pure water; basically by improving the conductivity of the water. The electrolytes do not participate in the overall reaction.

This activity introduces the concept of constant proportions; central to an understanding of compounds and chemical formulae. Hydrogen will form at the negative electrode and oxygen at the positive electrode. Students will notice more hydrogen gas being produced than oxygen gas. When students have produced about half a test tube of hydrogen gas they can do the 'pop test' for hydrogen.

Demonstrate to students that as they lift the test tube up vertically from the cup, the bicarb solution will fall out but the light hydrogen gas will remain trapped in the upside-down test tube. Light a match and rotate the test tube to bring the opening of the test tube near the flame. It will combust with a distinctive 'pop' sound.

Lesson plan

Step 1: Have the students set up the experiment using the instructions and images provided in the **Student Guide**. It is better (for inquiry-based teaching) that any explanations are given after students have observed the evolving gases at each electrode.

Step 2: If you have a set up using a voltmeter at the front of the class you may be able to evolve sufficient gases to demonstrate the 'pop-test' for hydrogen, and ignition of a glowing splint in oxygen.

Step 3: Discuss with the whole class, that everyone evolved more gas at the negative electrode than at the positive. At this stage make the point that when this experiment is carried out there is always twice as much hydrogen as oxygen produced. Now introduce that this proportion relates to the well-known formula for water.

Discussion:

1. What do you observe happening at each electrode?
2. Is there more gas produced at one of the electrodes? If so, which electrode?
3. Research the properties of hydrogen and oxygen. How are these different to the properties of water?



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3.6 Elements: the simplest substances

Curriculum links

ACSSU152 Differences between elements, compounds and mixtures can be described at a particle level.

KEY IDEAS

- An element is a pure substance that contains only one kind of atom.
- A compound is a pure substance that contains more than one kind of atom.

Lesson outcomes

At the end of this activity students will be able to:

- explain that there are a limited number of elements and that each element is made from just one type of atom
- understand that each element has its own symbol
- understand that each element has its own unique name, which matches the name of its constituent atoms
- classify elements into metals, non-metals and metalloids
- recognise that different elements melt at different temperatures.

What ideas might your students already have?

Students may be confused by elements that are made of more than one atom, such as the diatomic gases.

Key vocabulary:

Metalloid

Equipment list

Each GROUP will require:

- access to the **Student Digital**

Each STUDENT will require:

- *Science by Doing* **Notebook**

Things to consider and hints for success

Note that the elements are presented here in the periodic table, without any explanation of why they are presented in this particular way rather than as a straight list. It would be appropriate to draw students' attention to this and point out that the reason will be explored in the next activity.

The ideas presented in the *Student Guide* are very important and should be discussed at some length with the class both at the beginning and at the end of this activity.



Lesson plan

- Step 1:** Discuss the ideas presented in the *Student Guide*. It may be useful to have some samples of elements, such as carbon and sulfur and some pure metals to show.
- Step 2:** Students explore the interactive periodic table through the *Student Digital*. This could be done during class time or at home, with follow up discussion that next day.
- Step 3:** Students complete the **Notebook** activity.



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⚠ Always ensure safe and ethical disposal of chemicals.

⚠ Use teacher judgement when considering equipment substitutions.

⚠ Make considered choices selecting and using resources to minimise waste.

3.7 The purity of water

Curriculum links

ACSSU152 Differences between elements, compounds and mixtures can be described at a particle level.

KEY IDEAS

- An element is a pure substance that contains only one kind of atom.
- A compound is a pure substance that contains more than one kind of atom.
- In any particular compound atoms are always combined in the same proportion and are closely linked by chemical bonds.
- Mixtures contain different substances mixed together in varying proportions. They can be separated using a range of physical separation techniques.

Lesson outcomes

At the end of this activity students will be able to:

- identify that water is a compound; it is comprised of hydrogen and oxygen atoms always in the same proportion, which can be described using a chemical formula
- identify the different patterns of atoms in elements, compounds and mixtures
- recognise that both elements and compounds are classified as pure substances.

What ideas might your students already have?

Students misunderstand the meaning of the term 'pure'; they might refer to something that is clean and not contaminated.

Key vocabulary:

Element, compound, mixture, molecule.

Equipment list

Each GROUP will require:

- molecular model kit

Each STUDENT will require:

- *Science by Doing Notebook*

Things to consider and hints for success

Students find the concept of purity in this context very confusing, so it is useful to approach this chemical understanding in a very pragmatic way by emphasising the precise scientific definitions.



Teacher content information

A molecule is a substance containing two or more atoms in a fixed ratio, which are joined by chemical bonds e.g. O_2 , H_2O .

A compound is a substance made up of two or more different kinds of atoms e.g. H_2O is a compound, O_2 is not a compound.

Answers to the last part of the activity.

- | | |
|---|---------------------------|
| A. mixture (of compounds and molecules) | E. compound |
| B. element | F. molecule |
| C. mixture (of molecules) | G. mixture (of elements). |
| D. mixture (of molecules and elements) | |

Elements and compounds are all pure substances.

Lesson plan

Step 1: Discuss the information on the first page with the class (if you discussed compounds and element numbers at the end of the last activity this may be unnecessary).

Step 2: Part A: Identifying elements in compounds.

Students complete the last column of the table in the **e-Notebook** as described in the activity. Students share answers to ensure they are correct before moving on to Part B.

Step 3: Part B: Making molecular models.

In groups or individually, students complete the activity on the worksheet and answer the discussion questions in their **Notebooks**. Students share answers before moving on to Part C.

Step 4: The last part of this activity checks students' understanding of the elements, compounds and mixtures. This can be done individually for assessment, or in groups. Students look at diagrams A and B at the top of the page and record their answers to the question in their **Notebooks**. Share answers with the class to ensure students can identify the differences between elements, compounds and mixtures.

A: compound containing two elements (red and white)

B: mixture of two elements (red and white).

Suggested question:

What is different about these two substances?

Step 5: Students complete the activity by identifying the diagrams A-G as representing elements, molecules, compounds or mixtures.

Share student answers or check for assessment.

Suggested question:

Are they elements, molecules, compounds or mixtures?



3.8 The periodic table

Curriculum links

ACSSU152 Differences between elements, compounds and mixtures can be described at a particle level.

KEY IDEAS

- An element is a pure substance that contains only one kind of atom.

Lesson outcomes

At the end of this activity students will be able to:

- explain why the elements are set out in a periodic table.

What ideas might your students already have?

Students will come with no former understanding of the periodic table, which involves a range of complex concepts and ideas. They can be confused about whether the periodic table describes atoms or elements.

Key vocabulary:

Periodic table, periodicity

Equipment list

Each GROUP will require:

- access to **Student Digital**

Each STUDENT will require:

- *Science by Doing Notebook*

Things to consider and hints for success

The periodic table is one of the first, and the most difficult, idea in chemistry that students have to deal with. The complexity of its structure and the multiple scientific ideas involved provide enormous challenges to students.

In this light, this activity benefits from extended discussion with the class, so that they become familiar with the ideas presented in the **Student Guide**.

Teacher content information

Elements are listed in the periodic table in order of their atomic number; i.e. the number of protons in their atoms; the Atomic Number. The listed Atomic Weights are the average weight of the atoms found in the elements (on Earth), allowing for the different isotopes that are present. The rows are called 'periods' and the columns are called 'groups'. Some groups have specific names; such as the alkali metals, halogens, and noble gases.

Elements in the same column have similar properties which generally grade down the group. The metals in the middle group, the transition elements, all tend to have similar properties, which are also shared by the actinides and lanthanides.

Note that a great majority of elements are metals with overall similar appearance and generally similar properties (although reactivity varies a lot). An interesting observation is that elements tend to become denser towards the bottom of the table, because the atoms become more massive but do not change much in atomic volume. That is, elements at the bottom of the table are comprised of atoms that are significantly more massive but which do not take up much more space.



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▲ Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

▲ Always ensure safe and ethical disposal of chemicals.

▲ Use teacher judgement when considering equipment substitutions.

▲ Make considered choices selecting and using resources to minimise waste.

Lesson plan

- Step 1:** Discuss the ideas presented in the **Student Guide** with the class. Explain to them that the little *Student Digital* activity they will do matches Mendeleev's first work on producing the periodic table. Explain the importance of the periodic table and why it is called 'periodic'.
- Step 2:** Students complete the **Student Digital** activity and the **Notebook** activity. This could be done as homework.
- Step 3:** In follow up discussion remind students that the properties of substances has been used to build the periodic table.



3.9 Iron – element, compound and mixture

Curriculum links

ACSSU152 Differences between elements, compounds and mixtures can be described at a particle level.

KEY IDEAS

- An element is a pure substance that contains only one kind of atom.
- A compound is a pure substance that contains more than one kind of atom.
- In any particular compound atoms are always combined in the same proportion and are closely linked by chemical bonds.
- Mixtures contain different substances mixed together in varying proportions. They can be separated using a range of physical separation techniques.

Lesson outcomes

At the end of this activity students will be able to:

- list some compounds and mixtures of iron.

What ideas might your students already have?

Students may be aware that iron ore is an important commercial mineral mined in Australia, and that it rusts to produce a red colour. They may also know that iron is an important dietary mineral advertised in some breakfast cereals.

Key vocabulary:

Haemoglobin, chlorophyll, transpiration, anaemia, haemochromatosis, ore, ochre, alloy.

Equipment list

Each **GROUP** will require:

- access to **Student Digital**
- Resource Centre or library.

Each **STUDENT** will require:

- *Science by Doing* **Notebook**

Things to consider and hints for success

Use this activity to explore and broaden students' ideas about the nature and uses of iron by encouraging students to draw on knowledge from other subjects and everyday experiences.

Lesson plan

Step 1: A: Iron – an important element for life.

Students will need access to the internet or other reference sources. Individually or in groups, they will research how iron is important for animal and plant survival.

Step 2: B: Iron – a compound

Students will apply what they learnt in **Activity 3.6** to the different compounds of iron. They will also discover how Indigenous Australians used iron oxide. This activity links to **Activity 3.3**, where students found out about alloys.



Follow up:

Students could make their own ochre paint or you could invite a local Indigenous artist to show them how.

Making ochre paint

1. Combine 2 g of ochre with 5 mL of water in a small beaker or jar.
2. Add 1 tsp binder (see below) and stir well.
3. If too thick, add more water.
4. Paint a picture (on paper or rock).

Sources of ochre (or substitute): iron ore (grind in mortar and pestle), oxide pigment (from paint supplier), powdered paint, charcoal piece (grind in mortar and pestle) or powdered (paint supplier).

Binders: egg yolk or PVA.



3.10 Is it an element, compound or mixture?

Curriculum links

ACSSU152 Differences between elements, compounds and mixtures can be described at a particle level.

KEY IDEAS

- An element is a pure substance that contains only one kind of atom.
- A compound is a pure substance that contains more than one kind of atom.
- In any particular compound atoms are always combined in the same proportion and are closely linked by chemical bonds.
- Mixtures contain different substances mixed together in varying proportions. They can be separated using a range of physical separation techniques.

Lesson outcomes

At the end of this activity students will be able to:

- describe the differences between elements, compounds and mixtures, and identify examples of each.
- determine the name and number of elements in compounds.

What ideas might your students already have?

Students should be able to:

- define elements, compounds and mixtures
- describe the difference between compounds and mixtures
- identify examples of elements, compounds and mixtures
- identify the name and number of elements in a compound.

However, students may still find it difficult to distinguish between compounds and mixtures because the distinction between the two is somewhat mathematical.

Equipment list

Each STUDENT will require:

- *Science by Doing* **Notebook**
- **Activity Sheet 3.10** Elements, compounds and mixtures.

Things to consider and hints for success

This task could be used as either a **formative** or **summative assessment**. It could also be an open-book task, allowing students to use their **Notebooks** and **Student Guides**.

Lesson plan

Give each student a copy of **Activity sheet 3.10 Elements, compounds and mixtures (worksheet)**. They should individually complete this task for assessment.



4.1 Observing changes in water

Curriculum links

- ACSSU151 Properties of the different states of matter can be explained in terms of the motion and arrangement of particles.
KEY IDEAS
- All substances are made of particles, which are in constant motion. These particles are usually atoms and molecules.
 - The particle model can be used to explain a range of phenomena, including changes of state, and physical and chemical changes.
- ACSIS141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.
- ACSIS144 Construct and use a range of representations, including graphs, keys and models to represent and analyse patterns or relationships in data using digital technologies as appropriate.
- ACSIS148 Communicate ideas, findings and evidence-based solutions to problems using scientific language, and representations, using digital technologies as appropriate.

Lesson outcomes

At the end of this activity students will be able to:

- explain that substances can change state when they are heated or cooled.
- explain that a change in state does not produce a new chemical substance.
- interpret a graph documenting the change in temperature of water as it is heated and changes state.

What ideas might your students already have?

Students, along with most of the general public, will confuse water vapour and steam. Even the Oxford Dictionary refers to both an 'invisible vapour' and to a 'white mist' when defining steam. Students may also find it difficult to explain where condensed water comes from, or where the water goes when a surface dries.

Equipment list

Each GROUP will require:

- 250 mL beaker, half filled with crushed ice
- thermometer (-10 – 110°C) or digital thermometer
- retort stand and clamp
- stirring rod
- Bunsen burner, tripod, gauze mat and safety mat
- stopwatch
- access to **Student Digital**.



Each STUDENT will require:

- *Science by Doing Notebook*
- Safety glasses
- Graph paper

Things to consider and hints for success

Preliminary discussion with the class about what happens when water boils is important in this activity. It may be useful to explain directly that water vapour (i.e. the gas form) is an invisible and odourless gas.

When groups are formed, it is important to allocate students roles as timer, temperature reader and recorder. Make sure timers don't stop the stopwatch after recording each temperature in order to record later temperature changes on the minute. If it is too difficult to record each minute, the group should take a reading every two minutes.

Teacher content information

Most of us will refer to the white mist coming out of a boiling kettle as 'steam' in some contexts. However, it is important to keep in mind that from a scientific perspective it is important to distinguish clearly between the vapour state of water and the white mist formed from droplets of liquid water (or ice crystals in some situations).

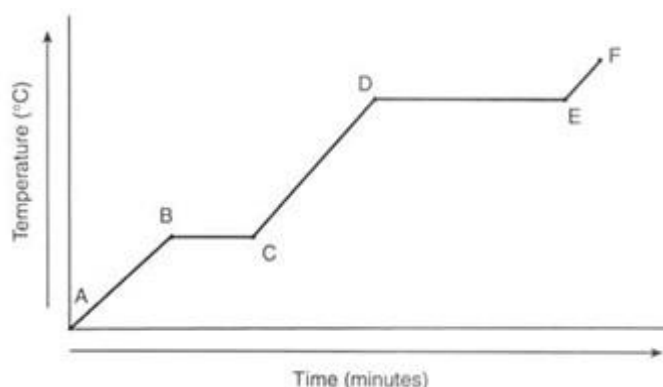
This activity also introduces the concept of a 'physical change'. This is a problematic concept. A lot of time and effort is often devoted to teaching students some distinction between chemical and physical changes to little educational effect. Change of state is usually given as one example of a physical rather than a chemical change, and is reasonably clear cut as an example. Other examples, however, are not so clear. For example, dissolving of salts in water is sometimes described as a physical change but also qualifies as a chemical change according to common definitions, since ionic bonds are clearly broken. In this case the dissolving of non-ionic compounds, such as sugar, in water is sometimes referred to as an example of a physical change, but this distinction is unlikely to help clarify the concept for students. Alloys (covered in **Activity 3.3**) are considered to be mixtures, and yet the structure of most alloys involves metallic bonding between the different metal atoms, so that the creation of an alloy could also be considered to be chemical in nature.

Other physical changes, such as bursting a balloon, tearing paper, breaking an egg are clearly macroscopic in nature and provide fairly trivial examples of the concept and do not help students understand what is happening in other situations at a molecular and atomic level. This activity focusses on change of state since this is an important phenomenon to understand in its own right, and not just as an example of a physical change.

Student graphs should look similar to the one shown below. Due to experimental error (inaccurate thermometer and students' recordings) graphs may not be exactly as shown, but should identify changes in state.

- A-B temperature of icy is rising – solid
- B-C temperature remains constant at around 0°C as solid becomes liquid (latent heat of fusion)
- C-D temperature of liquid rises – liquid
- D-E temperature remains constant at around 100°C as liquid becomes gas (latent heat of vaporization)
- E-F temperature of steam rises – gas (this section will not be observable in this activity since the gas disperses and re-condenses into mist).





Lesson plan

Step 1: Discuss the phenomenon of a boiling kettle with the class. Explain that water can exist as a vapour, but that this is a colourless and odourless gas.

Step 2: Groups form to conduct the experiment, collect results and answer Discussion questions.

Step 3: Students complete the Summary on page 49, individually or in groups.

Step 4: Students complete the digital activity and the **e-Notebook**

Discussion Questions

1. At what temperature did the ice start to melt?
2. At what temperature did the water start to boil?
3. On your graph label when the water was a solid, liquid, or a gas.

Further optional question (depending on whether it is appropriate for your class at this stage).

4. At the points on your graph where the graph was flat, the water was still being 'heated'. What do you think was happening to all that heat energy at these points?



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4.2 Ice play

Curriculum links

ACSSU151 Properties of the different states of matter can be explained in terms of the motion and arrangement of particles.

KEY IDEAS

- The particle model can be used to explain a range of phenomena, including changes of state, and physical and chemical changes.

ACSHE 135 Solutions to contemporary issues that are found using science and technology, may impact on other areas of society and may involve ethical considerations.

ACSHE136 People use science understanding and skills in their occupations and these have influenced the development of practices in areas of human activity.

Lesson outcomes

At the end of this activity students will be able to:

- explain how salt was used to lower the freezing point of water.

Equipment list

Each GROUP will require:

- 1 cup milk (or ½ cup of cream and ½ cup milk)
- 2 tablespoons sugar
- 1/4 teaspoon vanilla
- 6 cups of ice cubes (crushed)
- ½ cup table salt
- 1 small ziplock bag
- 1 large ziplock bag
- clean spoon for each student

Each STUDENT will require:

- Science by Doing Notebook*

Things to consider and hints for success

It is important that students are able to taste the ice-cream they produce. It is advised this activity is conducted using standard and clean kitchen equipment and not in a laboratory.

Teacher content information

Sodium chloride can depress the freezing point of water to around -21°C. Radiator water in cars, at least where they operate in cold climates, often has ethylene glycol added to the water. This can depress the freezing point to below -30°C. In the northern hemisphere the use of salt on roads in winter was often used lower the freezing point of ice on the roads. It is used less today because of the rusting effect on cars and adverse environmental effects.



Lesson plan

Step 1: Students follow the recipe for making ice cream. During the activity encourage conversations about changes of state, and what effect the salt is having on the melting point of ice.



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⚠ Make considered choices selecting and using resources to minimise waste.

4.3 Can matter skip states?

Curriculum links

ACSSU151 Properties of the different states of matter can be explained in terms of the motion and arrangement of particles.

KEY IDEAS

- The particle model can be used to explain a range of phenomena, including changes of state, and physical and chemical changes.

Lesson outcomes

At the end of this activity students will be able to:

- give examples and uses of sublimation.

What ideas might your students already have?

Students will not be aware that there is a change of state other than the usual ones.

Key vocabulary:

Sublimation

Equipment list

Each GROUP will require:

- access to **Student Digital**

Each STUDENT will require:

- Science by Doing* **Notebook**

Things to consider and hints for success

This activity relies on videos to provide experience of sublimation. Most substances used to demonstrate sublimation, such as dry ice, iodine and mothballs (naphthalene) and not able to be used by students at this level. You may choose to show some of these through a teacher demonstration.

Teacher content information

Whether or not sublimation occurs depends on a substance's triple point, the conditions of temperature and pressure at which the substance can coexist as solid, liquid and gas. Interestingly under very low pressure, for example, in outer space water ice will also sublime rather than melt.

Lesson plan

Step 1: Students complete the Digital activity and complete the **e-Notebook** questions.



4.4 Detecting an invisible gas

Curriculum links

- ACSSU151 Properties of the different states of matter can be explained in terms of the motion and arrangement of particles.
KEY IDEAS
- Different substances have different properties, allowing them to be used for different purposes.
- ACSSU152 Differences between elements, compounds and mixtures can be described at a particle level.
KEY IDEAS
- An element is a pure substance that contains only one kind of atom.
 - A compound is a pure substance that contains more than one kind of atom.
 - In any particular compound atoms are always combined in the same proportion and are closely linked by chemical bonds.

Lesson outcomes

At the end of this activity students will be able to:

- explain how carbon dioxide gas can be detected through its ability to smother a flame
- identify a range of gases and describe their properties.

What ideas might your students already have?

Students have difficulty recognizing gases as chemical substances. This is compounded by the fact that most gases that we might come across in our lives are colourless and odourless. They will also tend to classify gases as either 'air' (good) and 'gases' (bad).

Equipment list

Each GROUP will require:

- 1 large beaker
- 1 small beaker
- sodium bicarbonate
- hydrochloric acid (0.1 M)
- tea light candle
- spatula

Each STUDENT will require:

- *Science by Doing Notebook*

Teacher content information

The **Student Digital** activity introduces nine gases that students may come across during their lives. It is notable that every one of them is colourless and odourless, properties that make it difficult for students to develop a meaningful concept of what a gas actually is or to recognise their existence in the first place. These gases, however, have a wide range of chemical properties rendering some of them innocuous and other highly toxic. This raises considerable safety issues around their use.



Lesson plan

Step 1: Students complete the hands-on activity and answer the questions in their **Notebook**.

As an extension you could discuss with the class why this property of carbon dioxide (heavier than air) makes it dangerous in certain situations. Depending on the level of the class you could also discuss the reaction that occurred in more detail.

Step 2: Students complete the **Digital** activity and answer questions in the **e-Notebook**.



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⚠ Use teacher judgement when considering equipment substitutions.

⚠ Make considered choices selecting and using resources to minimise waste.

4.5 What is happening to me?

Curriculum links

ACSSU151 Properties of the different states of matter can be explained in terms of the motion and arrangement of particles.

KEY IDEAS

- The particle model can be used to explain a range of phenomena, including changes of state, and physical and chemical changes.

Lesson outcomes

At the end of this activity students will be able to:

- describe what happens to particles when heated and cooled.
- use correct terminology for what takes place when matter changes state.

Key vocabulary:

Melting, boiling, evaporation, condensation, freezing.

Equipment list

Each GROUP will require:

- a tray of sand
- polystyrene or plastic cups
- digital scales

Each STUDENT will require:

- Science by Doing* **Notebook**

Things to consider and hints for success

This task can be used as a **formative** or **summative assessment**. To assist students, especially those with low literacy, you could distribute a spelling list.

Step 1: Students describe what happens to a water molecule in an ice block left in the sun. Encourage them to be creative – write a story, draw a comic strip, make a poster etc.

You could allow students to use the information in their **Notebooks**, depending on the type of assessment you are undertaking. This could be a homework task.



5.1 Some chemical changes

Curriculum links

ACSSU225 Chemical change involves substances reacting to form new substances.

KEY IDEAS

- Chemical reactions involve the breaking down and creation of new pure substances. They involve chemical bonds being broken and reformed.

AC SIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.

Lesson outcomes

At the end of this activity students will be able to:

- describe a range of observable changes that occur during chemical reactions.

What ideas might your students already have?

Because the range of chemical reactions is so vast students have difficulty recognising any common theme. This is dependent on some level of understanding of what is happening at the molecular level.

Key vocabulary:

Chemical change

Equipment list

Each GROUP will require:

Test 1: Limewater test - test tube - drinking straw - lime water - test-tube rack	Test 2: Burning magnesium ribbon (Teacher demonstration ONLY) - magnesium ribbon - metal tongs - Bunsen burner - Evaporating basin - safety mat	Test 3: Starch test - test tube - starch in a dropper bottle - iodine in a dropper bottle - test-tube rack
Test 4: Steel wool and copper sulfate - test tube - copper sulfate solution - small piece of steel wool - test-tube rack	Test 5: Precipitation of lead iodide (Teacher demonstration ONLY) - small reagent bottle of 0.1 M potassium iodide - Dropper bottle with 0.01 M lead nitrate - Large test tube and test tube rack	



Each STUDENT will require:

- *Science by Doing Notebook*

Things to consider and hints for success

Understanding 'chemical change' as a unified concept requires consideration at the particle level. It is worth discussing the actual chemical rearrangements that have taken place with the class, either through word equations or balanced equations as appropriate. Each of the reactions is provided below. While balancing chemical equations is not part of this activity, students can still benefit from seeing the chemical equations and noting that the number of atoms before and after are the same, and that chemical reactions involve a rearrangement of the atoms.

According to Science Assist guidelines students are allowed to handle lead nitrate solution at levels below 0.5 M. However, make sure that students do not touch the tubes at the end or attempt to wash out the solutions or lead iodide precipitate themselves. This should be included in the *Risk Assessment* for this activity.

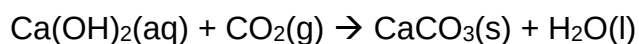
Disposal

Dispose starch solution after use.

Teacher content information

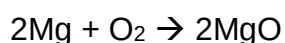
Test 1

Carbon dioxide passed into limewater gives a milky precipitate as calcium carbonate is formed



Test 2

Magnesium burnt in air produces a white ash. It is a highly exothermic reaction.

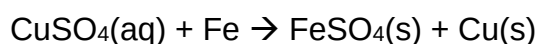


Test 3

When starch is mixed with iodine in water (or a potassium iodide solution), an intensely coloured starch/iodine complex is formed. This reaction is not fully understood. Somehow the iodine ions get stuck in the some of the coiled sections of the starch molecules. It is probably somewhat academic as to whether this is a chemical or a physical change, depending on how the iodine is actually bonded to the starch, but it is a clearly observable, and commonly used, reaction.

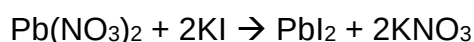
Test 4

This is a displacement reaction, where the copper comes out of solution and displaces the iron. Because the iron sulfate is not very soluble a rusty coloured precipitate might be observed.



Test 5

This is a straight precipitation reaction, producing the colourful (yellow) lead iodide precipitate. It is worth pointing out to students that lead salts used to be commonly used in paints, but were toxic.



Lesson plan

- Step 1:** Explain to students that they will perform a series of tests involving chemical reactions and will record their observations in their Notebooks. They should first copy the table in the **Student Guide** into their **Notebooks**.
- Step 2:** Groups conduct each test and record observations. Show Test 2 and Test 5 to the class.
- Step 3:** When testing is completed, students will answer the discussion questions in their Notebooks, and share their observations with the rest of the class. They may like to include the chemical reactions in their Notebooks.



5.2 Observing reactions

Curriculum links

ACSSU225 Chemical change involves substances reacting to form new substances.

KEY IDEAS

- Chemical reactions involve the breaking down and creation of new pure substances. They involve chemical bonds being broken and reformed.

AC SIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.

Lesson outcomes

At the end of this activity students will be able to:

- describe a range of observations that provide evidence for chemical changes.

What ideas might your students already have?

Students know that a new substance is formed but the range of observations that might indicate this are so varied that they will still be confused about the unified concept of 'chemical change'.

A 'chemist's' view is likely to revolve around an understanding of molecular rearrangements and bonding and the latter is beyond the scope of this unit.

Key vocabulary:

Precipitate, reaction

Equipment list

Each GROUP will require:

Test 1 250 mL beaker - copper strip - zinc strip - copper sulfate solution 2 electric leads with alligator clips - LED (light emitting diode)	Test 2 - test tube - hydrochloric acid in a dropper bottle - small piece of magnesium ribbon	Test 3 (Teacher demonstration ONLY) - test tube - dropper bottle of lead nitrate - dropper bottle of potassium iodide
Test 4 - test tube - vinegar in a dropper bottle - sodium hydrogen carbonate - popstick	Test 5 - test tube - copper sulfate solution - sodium hydroxide solution	



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▲ Always ensure safe and ethical disposal of chemicals.

▲ Use teacher judgement when considering equipment substitutions.

▲ Make considered choices selecting and using resources to minimise waste.

Each STUDENT will require:

- *Science by Doing Notebook*
- safety glasses

Things to consider and hints for success

It is easiest to give each group a complete set of equipment to reduce movement around the room and to ensure students can find all they need.

Teacher content information

Evidence for chemical changes:

1. produces electricity (**Test 1**)
2. produces a gas (**Test 2**)
3. produces a colour change (**Test 3**)
4. produces a change in temperature (**Test 4** – exothermic)
5. produces a precipitate (**Tests 5 and 3**)

Lesson Plan

Step 1: Explain students will conduct a series of tests and record their observations. They should first copy the table in the *Student Guide* into their **Notebooks**.

Step 2: Groups conduct Tests 1, 2, 4 & 5, teacher demonstrates Test 3, and students record their observations.

Step 3: Discuss students' observations for each test and identify the evidence demonstrated. Students record evidence in their **Notebooks**.

Discussion

How can you tell if a chemical change has taken place?



5.3 Reactants and products

Curriculum links

ACSSU225 Chemical change involves substances reacting to form new substances.

KEY IDEAS

- Chemical reactions involve the breaking down and creation of new pure substances. They involve chemical bonds being broken and reformed.

AC SIS140 Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed.

Lesson outcomes

At the end of this activity students will be able to:

- identify the reactants and products in a chemical reaction
- write the word equations for chemical reactions.

What ideas might your students already have?

Students may have been introduced to word equations for chemical reactions in earlier activities. They may not have extended this thinking in terms of reactants and products; i.e. a before and after point of view.

Key vocabulary:

Reactant, product

Equipment list

Each GROUP will require:

Pop test:

- test tube
- hydrochloric acid in a dropper bottle
- small piece of magnesium ribbon
- matches

Test 1

- test tube
- hydrochloric acid in a dropper bottle
- calcium carbonate

Test 2

- test tube
- hydrochloric acid in a dropper bottle
- granulated zinc

Test 3

- test tube
- cobalt chloride solution in a dropper bottle
- sodium carbonate solution in a dropper bottle

Each STUDENT will require:

- Science by Doing Notebook*



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Wearing appropriate personal protective equipment is advised, over and above what is stated as safety gear.

Always ensure safe and ethical disposal of chemicals.

Use teacher judgement when considering equipment substitutions.

Make considered choices selecting and using resources to minimise waste.

Things to consider and hints for success

It is easiest to give each group a complete set of equipment to reduce movement around the room and to ensure students can find all they need.

Teacher content information

Pop test

magnesium (s) + hydrochloric acid (aq) → magnesium chloride (aq) + hydrogen (g)

Test 1

calcium carbonate (s) + hydrochloric acid (aq) → calcium chloride (aq) + water (l) + carbon dioxide (g)

Test 2

zinc (s) + hydrochloric acid (aq) → zinc chloride (aq) + hydrogen (g)

Test 3

Cobalt chloride (aq) + sodium carbonate (l) → sodium chloride (aq) + cobalt carbonate (s)

Lesson Plan

Step 1: Students read the information at the beginning of the activity, conduct the first experiment and complete the discussion questions.

Step 2: Discuss students' answers to ensure they have the correct understanding of products and reactants.

Step 3: Students complete the remaining three tests and discussion questions.

Step 4: Students share results with the class.



5.4 Some natural reactions

Curriculum links

ACSSU225 Chemical change involves substances reacting to form new substances.

KEY IDEAS

- Chemical reactions involve the breaking down and creation of new pure substances. They involve chemical bonds being broken and reformed.

Lesson outcomes

At the end of this activity students will be able to:

- identify some naturally occurring reactions.
- explain why photosynthesis is one of the most significant chemical reactions.

What ideas might your students already have?

Students will be aware of photosynthesis through previous units in biology, but may not appreciate its chemical nature, or its importance.

Key vocabulary:

Photosynthesis

Equipment list

Each GROUP will require:

- access to **Student Digital**

Each STUDENT will require:

- *Science by Doing* **Notebook**

Things to consider and hints for success

To ensure students are on track, you could discuss their responses after each **Notebooking** task, rather than at the end of the activity. If they work in **Learning Partners** when using the *Student Digital*, you could ask one partner to work on the first **Notebooking** task (photosynthesis) and the other to work on the second (corrosion). Alternatively, half the class could complete the first task and the other half, the second. They then share responses.

Teacher content information

Oxygen is highly reactive and cannot survive long in the atmosphere without reacting with other chemicals. Its presence in the Earth's atmosphere is totally dependent on its replacement through photosynthesis, a highly energy demanding reaction powered by light. All the oxygen currently in the atmosphere came from photosynthesis. Astronomers believe that if oxygen is ever detected in the atmosphere of another planet, probably through spectral analysis, it will provide strong evidence for the existence of life on that planet.

Lesson Plan

Step 1: Explain students will investigate some natural chemical reactions.

Step 2: Students work through the activity, completing all questions in their **Notebooks**.

Step 3: Discuss student responses as a class.



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▲ Make considered choices selecting and using resources to minimise waste.

5.5 Investigating some useful reactions

Curriculum links

ACSSU225 Chemical change involves substances reacting to form new substances.

KEY IDEAS

- Chemical reactions involve the breaking down and creation of new pure substances. They involve chemical bonds being broken and reformed.

ACSHE 134 Scientific knowledge has changed peoples' understanding of the world and is refined as new evidence becomes available.

ACSHE136 People use science understanding and skills in their occupations and these have influenced the development of practices in areas of human activity.

AC SIS146 Reflect on scientific investigations including evaluating the quality of the data collected, and identifying improvements.

Lesson outcomes

At the end of this activity students will be able to:

- give an example of a useful reaction and describe the properties that make it so.

Key vocabulary:

Synthetic, plastic.

Equipment list

Each GROUP will require:

Investigation 1:	Investigation 2:	Investigation 3:
10 mL of latex 10 mL syringe 100 mL beaker - glass stirring rod - Pasteur pipette 2M hydrochloric acid - metre rule	50 mL of milk 10 mL vinegar 10mL measuring cylinder 50 mL measuring cylinder 250 mL beaker - thermometer - water bath set to 50-70 °C - glass dropper - stirring rod - filter funnel - filter-funnel stand - coffee-filter paper - paper towel	- two 250 mL beakers 2 teaspoons - stirring rod - filter funnel - filter-funnel stand - coffee-filter paper 2 tablespoons powdered non-fat milk 20 mL vinegar - sodium hydrogen carbonate - set of measuring spoons - measuring cup - hot water 2 small pieces of paper 2 wooden pop sticks



Each STUDENT will require:

- *Science by Doing Notebook*
- safety glasses

Things to consider and hints for success

Students can do one or more investigations. They could compete to see whose rubber ball bounces the highest or test the ball in **Investigation 1** against the ball in **Investigation 2**. It is easiest to give each group a complete set of equipment for each investigation, to reduce movement around the room and ensure students can find all they need. Students could first complete a *Risk Assessment*.

Teacher content information

Whether or not sublimation occurs depends on a substance's triple point, the conditions of temperature and pressure at which the substance can coexist as solid, liquid and gas. Interestingly under very low pressure, for example, in outer space water ice will also sublime rather than melt.

Lesson Plan

Step 1: Students complete each investigation and answer discussion questions in their *Notebooks*.

Step 2: Students complete the *Digital activity* and the related *Notebook* exercise.

Step 3: Share student responses with the class.



5.6 A rusty problem

Curriculum links

ACSSU225 Chemical change involves substances reacting to form new substances.

KEY IDEAS

- Chemical reactions involve the breaking down and creation of new pure substances. They involve chemical bonds being broken and reformed.

AC SIS141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.

Lesson outcomes

At the end of this activity students will be able to:

- define rust and what causes it.

What ideas might your students already have?

Students will be aware of rust in some contexts but will not be aware of its chemical nature.

Key vocabulary:

Rust

Equipment list

Each GROUP will require:

- 4 test tubes
- 1 stopper to fit test tube
- test-tube rack
- calcium chloride
- 4 small pieces of steel wool
- boiled water
- vegetable oil

Each STUDENT will require:

- *Science by Doing Notebook*
- safety glasses

Things to consider and hints for success

Students could use thumbs up/thumbs down to indicate their predictions e.g. 'Do you agree with Katie?'

Group members may disagree.

It is easiest to give each group a full set of equipment, to reduce movement around the room and ensure all students find what they need. Students could first complete a *Risk Assessment*.



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This experiment must be observed and results recorded over four or five days. It should start early in the week, with observations made each lesson. Meanwhile, students can move on to the investigation of the effect of salt on rusting.

Teacher content information

Rusting involves the oxidation of iron (Fe). The products are generally some form of hydrated iron oxide ($\text{Fe}_2\text{O}_3 \cdot n\text{H}_2\text{O}$) and hydroxides of iron III. Hence rusting occurs most vigorously in environments rich in oxygen and water, which acts as a catalyst. The presence of salt (NaCl) can speed up the process of rusting as it enhances the electrochemical reactions taking place.

As rust is produced it tends to expand, possibly causing severe damage to the surrounding structures, and forms a flaky and friable layer. Unlike what happens when several other metals corrode (including copper, zinc and aluminium) the surface layer of rust does not tend to protect the underlying layers, so that rusting can continue until the whole iron structure totally disintegrates.

Lesson Plan

Step 1: Ask: 'Where have you seen rust?' Students record answers in their **Notebooks** or you can write them on the board.

Step 2: Ask students to consider a selection of rusty objects (if available) and the images in their *Student Guide*. Ask what they have in common.

Step 3: Ask students to consider what causes rust. A list of suggestions may be used to prompt ideas and discussion.

Step 4: Explain that students will conduct an investigation to identify what causes rusting and to test their predictions.

Step 5: Groups conduct the investigation, record their observations and answer the discussion questions.

Step 6: Students compare their results to those of Katie, Joshua, Olivia and Iggy, and complete the discussion questions.

Step 7: Discuss students' results and the cause of rust. Ask them to complete the question at the end of the activity.

Step 8: Students plan and conduct their experiment to test how salt affects the rusting reaction.

Follow up:

Students could investigate rusting in steel wool by varying its exposure to air and water.



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5.7 Preventing rust

Curriculum links

ACSSU225 Chemical change involves substances reacting to form new substances.

KEY IDEAS

- Chemical reactions involve the breaking down and creation of new pure substances. They involve chemical bonds being broken and reformed.

ACSHE 135 Solutions to contemporary issues that are found using science and technology, may impact on other areas of society and may involve ethical considerations.

ACSI141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.

Lesson outcomes

At the end of this activity students will be able to:

- describe some methods of preventing corrosion of iron.

What ideas might your students already have?

Students will be aware, from **Activity 5.2**, that electricity can be involved in chemical reactions.

Key vocabulary:

Corrosion, control, sacrificial anode.

Equipment list

Each GROUP will require:

- 6 test tubes
- test-tube rack
- salt solution
- 4 iron nails
- 1 nail painted with rust-proof paint
- 1 galvanised nail
- vaseline
- copper wire
- magnesium ribbon

Each STUDENT will require:

- *Science by Doing Notebook*

Things to consider and hints for success

This experiment must be left for at least three days. Students can complete all or some of the questions. They could research the questions at the end of the activity while they await their results.



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Suggested research questions:

How do sacrificial anodes work?

How is corrosion managed for the Sydney Harbour Bridge (or a similar large steel structure in your state or territory)?

Teacher content information

Whether or not sublimation occurs depends on a substance's triple point, the conditions of temperature and pressure at which the substance can coexist as solid, liquid and gas. Interestingly under very low pressure, for example, in outer space water ice will also sublime rather than melt.

Lesson Plan

Step 1: Explain students will investigate different ways to prevent corrosion of iron.

Step 2: Groups set up their investigations.

Step 3: Students make and record observations in their **Notebooks** over three days.

Step 4: Students complete the discussion questions and research the final two questions if necessary.



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5.8 Do other metals corrode like iron?

Curriculum links

ACSSU225 Chemical change involves substances reacting to form new substances.

KEY IDEAS

- Chemical reactions involve the breaking down and creation of new pure substances. They involve chemical bonds being broken and reformed.

AC SIS140 Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed.

AC SIS141 Measure and control variables, select equipment appropriate to the task and collect data with accuracy.

AC SIS145 Summarise data, from students' own investigations and secondary sources, and use scientific understanding to identify relationships and draw conclusions based on evidence.

Lesson outcomes

At the end of this activity students will be able to:

- demonstrate their ability to design and conduct an investigation to answer a question.

Equipment list

Each GROUP will require:

- single strips of metals such as aluminium, copper, iron, lead, magnesium and zinc
- steel wool (or emery paper)
- students select their own equipment.

Each STUDENT will require:

- Science by Doing Notebook*

Things to consider and hints for success

Students design and conduct an investigation. If they have not done this previously or recently, you could use this investigation to model the process. This activity could also be used for assessment. Try to get strips of metal of similar sizes. You don't have to provide all those suggested. Students should first complete a *Risk Assessment*.

Teacher content information

Whether or not sublimation occurs depends on a substance's triple point, the conditions of temperature and pressure at which the substance can coexist as solid, liquid and gas. Interestingly under very low pressure, for example, in outer space water ice will also sublime rather than melt.

Lesson Plan

Step 1: Explain to students that they will design and conduct an investigation to investigate the question 'Do other metals corrode like iron?' Explain that they will be provided with strips of different metals, but must decide on their own equipment.



5.9 Chemical hazards

Curriculum links

- ACSHE 134 Scientific knowledge has changed peoples' understanding of the world and is refined as new evidence becomes available.
- ACSHE 135 Solutions to contemporary issues that are found using science and technology, may impact on other areas of society and may involve ethical considerations.
- AC SIS140 Collaboratively and individually plan and conduct a range of investigation types, including fieldwork and experiments, ensuring safety and ethical guidelines are followed.

Lesson outcomes

At the end of this activity students will be able to:

- describe some safety precautions required when in danger of being exposed to hazardous particles in the air.

Equipment list

The **CLASS** will require:

- computer access
- internet access
- Student Digital*

Each **STUDENT** will require:

- Science by Doing Notebook*

Things to consider and hints for success

This activity explores safety precautions for hazardous particles in the air. It can be done at any time in the unit.

As you teach:

Step 1: Students work through the activity, answering the questions in their **Notebooks**.



5.10 Sample test

A sample summative test (editable version) and marking scheme are available to teachers by email request from *Science by Doing* (sbd@science.org.au). In your email request please include your school (or university) and contact details.

Note - *Science by Doing* provides sample assessment items and whilst every effort has been made, the security of these items cannot be guaranteed. Science by Doing encourages teachers to modify the items to suit individual teaching programs.



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